



Molecular analysis of streptomycin-resistance associating genes in *Mycobacterium tuberculosis* isolates from Nepal

Dipti Shrestha^a, Bhagwan Maharjan^b, Nan Aye Thida Oo^{a,c}, Norikazu Isoda^{d,e},
Chie Nakajima^{a,d}, Yasuhiko Suzuki^{a,d,*}

^a Division of Bioresources, Hokkaido University Research Center for Zoonosis Control, Kita 20, Nishi 10, Kita-ku, Sapporo, 001-0020, Japan

^b German Nepal Tuberculosis Project c/o Nepal Anti-Tuberculosis Association, Kalimati, Kathmandu, Post Box 1494, Nepal

^c Bacteriology Research Division, Department of Medical Research, Ministry of Health and Sports, Myanmar

^d The Global Station for Zoonosis Control, Global Institution for Collaborative Research and Education (GI-Core), Hokkaido University, Kita 20, Nishi 10, Kita-ku, Sapporo, 001-0020, Japan

^e Unit of Risk Analysis and Management, Hokkaido University Research Center for Zoonosis Control, Kita 20, Nishi 10, Kita-ku, Sapporo, 001-0020, Japan

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ABSTRACT

Mutation in *rpsL* (encoding ribosomal protein S12), *rrs* (encoding 16S ribosomal RNA) and *gidB* (encoding 7-methylguanosine methyltransferase) are associated with resistance to streptomycin (STR), which is used for the treatment of multi-drug resistant tuberculosis (MDR-TB) in Nepal. The aim of our study is to analyze the correlation between mutations in the target genes and STR-resistance in 197 *Mycobacterium tuberculosis* (MTB) isolates from Nepal. Mutations in *rpsL* was harbored by 65.9% of isolates, in which the most common mutation in *rpsL* is caused by K43R (58.8%) and were significantly associated with Beijing genotype ($P < 0.001$). About 13.2% of isolates harbored mutations in two highly mutable regions of *rrs*, the 530 loop and the 912 region. About 13.2% of *gidB* mutants do not show any mutation in *rpsL* and *rrs*, which might suggest the role of *gidB* mutations in STR-resistance in MTB. In addition, 5.6% of isolates do not show any mutations in three genes examined, suggesting the involvement of other mechanism in STR-resistance in MTB. Our findings can be implemented for the establishment of molecular STR-susceptibility testing, in which tuberculosis can be treated with appropriate drugs and can improve control strategies for DR-TB.

1. Introduction

Tuberculosis (TB), caused by *Mycobacterium tuberculosis* (MTB), remains a major global health problem. According to the Global Tuberculosis Report (2019), World Health Organization (WHO) estimated 10.0 million cases and 1.5 million deaths in 2018 [1]. Drug-resistant TB (DR-TB) tends to increase worldwide, which causes difficulties in TB treatment. Multi-drug resistant TB (MDR-TB), which is defined as TB resistant to at least two potent anti-TB drugs, isoniazid (INH) and rifampicin (RIF), is a serious threat to fight against TB [2]. The WHO estimated that, worldwide in 2018, there were half a million new cases of rifampicin-resistant TB (RR-TB), and of this, 78% had MDR-TB [1]. Almost half (50%) of the global TB cases were in India (27%), China (14%), and the Russian Federation (9%).

National TB Program (NTP) Nepal registered 32,474 TB cases in 2018 and estimated annual case notification rate of 112/100,000 [3].

However, WHO estimated incidence rate of 151/100,000 and death rate of 19/100,000 for Nepal, meaning NTP Nepal is yet to identify the missing cases in the community and bring them to treatment [1].

Streptomycin (STR) has been an important drug for the treatment of pulmonary TB in human [4]. For the retreatment TB patients in Nepal, STR has been included in intensive phase for two months with other first line anti-TB drugs [5]. However, NTP no longer recommends STR to be used in retreatment regimen in Nepal [3]. According to the WHO recommended regimen as published in “WHO treatment guidelines for MDR and RR-TB, 2018 Pre-final test” [6] and updated guidelines in “WHO consolidated guidelines on drug-resistant tuberculosis treatment, 2019” [7], STR is categorized into Group C agents in which it may be included in the treatment of MDR/RR-TB patients aged 18 years or more on longer regimens when susceptibility has been demonstrated and adequate measures to monitor for adverse reaction can be ensured, if amikacin is not available [3]. To move into the new WHO

* Corresponding author. Division of Bioresources, Hokkaido University, Research Center for Zoonosis Control, Kita 20, Nishi 10, Kita-ku, Sapporo, 001-0020, Japan
E-mail address: suzuki@czc.hokudai.ac.jp (Y. Suzuki).

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recommendation, NTP will develop a comprehensive transition plan [3]. With an increment of MDR-TB cases, and in addition of MDR-TB isolates resistant to second-line drugs and newly discovered drugs, STR should be used in combination with other drugs for MDR-TB treatment regimen. Despite its high rate of side effects and risk of developing drug resistance in TB of its single use in a regimen [4], it is crucial to understand the molecular mechanism of resistance to STR in MTB from Nepal.

STR impairs translational proofreading, leading to the blockade of protein synthesis. Resistance to STR is associated with mutations in *rpsL* and *rrs* genes, encoding for ribosomal protein S12 and 16S ribosomal RNA, respectively. The most frequent mutations localize around two regions of *rrs*, namely the 530 loop and the 912 region, or at codons 43 or 88 within *rpsL* [8,9]. Mutations within the *gidB* gene, encoding a conserved 7-methylguanosine methyltransferase specific for the 16S rRNA, have been reported to confer low-level STR-resistance in MTB [10]. Since, STR-resistance associating genes vary geographically, it is important to perform the molecular studies which will detect the common mutations in target genes conferring STR-resistance among MTB strains [10,11].

The time consuming conventional proportional method using Löwenstein-Jensen medium is the only available drug susceptibility testing (DST) for STR in Nepal. Recently, NTP have adopted the latest diagnostic tools such as Xpert MTB/RIF, Line probe assay (LPA), and Mycobacterium Growth Indicator Tube (MGIT) [3]. Despite the high number of MDR-TB cases with STR-resistant MTB strains, no genetic information regarding the molecular mechanism of STR-resistant MTB in Nepal is currently available.

The main aim of this study is to characterize the molecular mechanism of STR-resistance of MTB in Nepal; to analyze the correlation between mutations in the *rpsL*, *rrs* and *gidB* and STR-resistant MTB isolates from Nepal; and to study the association of those mutations with different MTB genotypes. To elucidate mutations responsible for the resistance to STR, we performed sequence analysis of STR-resistance associating genes (*rpsL*, *rrs* and *gidB*) in MDR-MTB isolates from Nepal.

2. Materials and methods

2.1. Studied sample

MDR-TB isolates with resistant to all four first line drugs INH, RIF, STR and ethambutol (EMB) has been preserved at the German Nepal Tuberculosis Project (GENETUP), Nepal. A total of 197 representative STR-resistant MTB isolates from April 2009 to March 2012 were selected in this study. These samples were subjected to genotyping and mutation analysis.

2.2. Drug susceptibility testing (DST)

DST for INH, RIF, EMB (FatolArzneimittel GmbH, Schiffweiler, Germany) and STR (Sigma-Aldrich, St. Louis, MO) were performed by the indirect proportional method on Löwenstein-Jensen medium with the critical concentrations of 0.2, 40, 2 and 4 µg/ml, respectively [12, 13].

2.3. DNA extraction

Extraction of DNA was done for PCR following the manufacturers protocol using the Genotype MTBDRplusVer2 (HainLifescienceNehren, Germany) [14].

2.4. Spoligotyping

The genotypes of the MTB isolates were identified using spoligotyping which was performed as described previously [15,16].

2.5. PCR amplification and sequencing

Amplification of the 437 bp, 590 bp and 870 bp fragments of *rpsL*, *rrs* and *gidB*, respectively, were performed using primers shown in Table 1. PCRs were carried out in a thermal cycler (Bio-Rad Laboratories, Inc., Hercules, CA) under the condition previously described [17]. The presence of PCR products was confirmed by agarose gel electrophoresis. Purified PCR products were sequenced using DNA sequencer, 3500 Genetic Analyzer (Thermo Fisher Scientific, Inc., Waltham, MA) using both forward and reverse primers same as for amplification. The obtained sequences were aligned with the corresponding sequences from the wild-type *M. tuberculosis* H37Rv using Bio-Edit software (version 7.0.9) [18].

2.6. Statistical analysis

All data were analyzed using R version 3.6. Pearson's Chi-square test was used to compare different variances and to determine the association between MTB lineages, drug-resistance profile and mutations in the genes examined. A p-value less than 0.05 was considered statistically significant.

Ethical approval

This study proposal was submitted to Nepal Health Research Council (NHRC) with register no 136/2013 on Sep 20, 2013 and approved by Ethical Review Board on Nov 29, 2013.

3. Results

3.1. Mycobacterium tuberculosis genotypes

Through spoligotyping, the genotypes of 197 STR-resistant MDR-MTB isolates were detected as 11 lineage 1, 116 lineage 2, 47 lineage 3 and 23 lineage 4 (Table 2).

3.2. Mutation in *rpsL*, *rrs* and *gidB*

Mutations in only *rpsL* were harbored by 65.9% (130/197) of isolates. Mutations conferring amino acid substitution at *rpsL* codon 43 and 88 were found in 58.8% (116/197) and 7.1% (14/197) of isolates, respectively (Table 3).

There were seven different substitutions found at seven different positions in the *rrs* gene. Mutations in only *rrs* was harbored by 13.2% (26/197) of isolates (Table 3). Mutations at positions A514C and C517T of *rrs* were found in 5.6% (11/197) and 2.5% (5/197) of isolates, respectively. Mutations around 912 loop (A906G, A907C, A908C or A912C) of *rrs* were found in 5.1% (10/197) of isolates. A co-mutation in *rpsL* and *rrs* (*rpsL* K43R, *rrs* A907C; *rpsL* K88R, *rrs* A644G; and *rpsL* G118D, *rrs* A514C) was found in 1.5% (3/197) of isolates.

Mutations in *gidB* was harbored by 25.8% (51/197) of isolates (Table 5). The *gidB* mutants showing mutations in *rpsL* and/or *rrs* were found in 12.7% (25/197) of STR-resistant isolates (Supplementary Table). About 13.2% (26/197) of *gidB* mutants do not show any

Table 1

Primers used for PCR amplification and sequencing of streptomycin-resistance associated genes in *M. tuberculosis*.

Locus	Primers	Nucleotide sequence (5'-3')	Product size (bp)
<i>rpsL</i>	Forward (ON-1505)	GGCCGACAAACAGAACGT	437
	Reverse (ON-1506)	CTTCACCACTGGGTGAC	
<i>rrs</i>	Forward (ON-1507)	GATGACGGCCTTCGGTTGT	590
	Reverse (ON-1508)	AGGCCACAAGGGAACGCCTA	
<i>gidB</i>	Forward (ON-1509)	CGCCGAGTCGTTGTGCT	870
	Reverse (ON-1510)	AGCCTGCCCCGACCGTTA	

bp represents base pair.

Table 2Distribution of 197 streptomycin-resistant *Mycobacterium tuberculosis* isolates into different lineages.

Lineages	Number of isolates
1	11
2 (Beijing)	116
3	47
4	23
Total	197

mutation in *rpsL* and *rrs* (Supplementary Table). No mutations in three genes examined in this study was found in 5.6% (11/197) of isolates (Supplementary Table).

4. Discussion

According to our knowledge, this is the first study on the molecular characterization conferring STR-resistance associating genes in MTB isolates from Nepal. It has been determined that mutations in *rpsL* and *rrs* genes are associated with STR-resistance in MTB [20]. In MDR-TB isolates, about 80.7% (159/197) of STR-resistant MTB isolates harbored mutations in either *rpsL* or *rrs*. This is within the range, reported from Asian countries, 40%–98% [19–23]. Mutations in *rpsL* was harbored by 65.9% (130/197) of isolates (Table 3), which suggest that these mutations strongly correlate to STR-resistance in MTB. The most common mutation in *rpsL* codon K43R was harbored by 58.8% of *rpsL* mutants in this study. The detection rate of mutations in *rpsL* codon K43R varies considerably between geographical areas (Table 4). The high detection rate of mutation in *rpsL* K43R in Nepal might be due to its non-restrictive mutation leading to no fitness cost in MTB [24] which can be widely transmitted; and the bad practice of STR mono-therapy usage for the treatment of TB and other bacterial diseases in the past in Nepal [25]. The predominant mutation *rpsL* K43R was found in 80%

(92/117) of Beijing isolates. There was a significant association between Beijing strains and mutation K43R compared with the group of STR-resistant non-Beijing isolates ($P < 0.001$). Sun et al. reported that mutation in *rpsL* causing K43R showed strong association with Beijing genotype in STR-resistance isolates [22]. However, in India and Mexico, where Beijing genotype was not frequent, mutations in these two loci had no significant contribution to Beijing genotype [19,21]. The Beijing genotype has been reported as more virulent than other MTB genotypes, and has a propensity to acquire the *rpsL* K43R mutation [22]. The high mutation rate in *rpsL* K43R represents the geographical areas predominate by Beijing genotype family. In contrast, mutation *rpsL* K88R was not significantly associated with the Beijing family ($P = 0.34$). Mutations in *rpsL* K43R and G71D were found in one STR-resistant non-Beijing isolate. As mutation K43R of *rpsL* strongly contributes to STR-resistance in MTB, it is doubtful that whether mutation G71D of *rpsL* contribute to STR-resistance in MTB or not. Mutation conferring G71D found in this study, was reported by study [26] with an isolate with mutation in *rpsL* G71S (GGC→AGC), compounded with mutation in *rpsL* K43R; and might be of only minor importance.

About 13.2% (26/197) of isolates harbored *rrs* mutations (except co-mutation in *rpsL*) in our study, which showed a frequency range of 0–12.5% (Table 4) in the Mideast and Southeast Asian countries [27–31]. Mutations in *rrs* were found in two highly mutable regions of *rrs*, the 530 loop and the 912 region, which suggest these mutations contributed to STR-resistance in MTB as reported [8,10,19–21,27,32–36].

Three strains showed co-mutations in both *rpsL* and *rrs*. Among them, one strain showed mutations in *rpsL* causing K43R and *rrs* 912 region A907C. Meier et al. had reported the similar findings of co-mutations in *rpsL* and *rrs*, which described that the *rrs* mutation (A→G at position 904) presumably conferred as yet ill-defined partial resistance to STR in vivo, thereby increasing the number of cells which could acquire a second mutation in the *rpsL* (alteration of K88), conferring resistance to STR-induced misreading [33]. Similar co-mutations *rpsL* K43R/*rrs*

Table 3Distribution of mutations in *rpsL* and *rrs* genes among different genotypes of 197 streptomycin-resistant *Mycobacterium tuberculosis* isolates.

<i>rpsL</i> nucleotide (amino acid substitution)	<i>rrs</i> nucleotide	Lineage 1	Lineage 2	Lineage 3	Lineage 4	No. of isolates
A128G (K43R)	–	2	91	13	9	115
A128G (K43R)	A907C		1			1
A128G (K43R), G212A (G71D)*	–			1		1
A263G (K88R)	–		12	2		14
A263G (K88R)	A644G*		1			1
G353A (G118D)*	A514C				1	1
–	A514C	1	5	2	3	11
–	A517T	1	1	2	1	5
–	A906C			1		1
–	A907C		3			3
–	A908C			3	2	5
–	A912G			1		1
–	–	7	2	22	7	38
Total		11	116	47	23	197

Fs denotes frameshift mutation.

No. represents number.

* denotes mutation previously not reported.

- denotes no mutation.

Table 4Detection rate of mutations in *rpsL*, *rrs* and *gidB* between countries.

Countries	This study (N = 197)	India (N = 52)	Pakistan (N = 32)	China (N = 227)	Myanmar (N = 141)	Thailand (N = 46)
Mutations						
<i>rpsL</i> codon K43R	58.8%	32.6%	34.3%	47.5%	64.0%	69.5%
<i>rpsL</i> codon K88R	7.1%	19.2%	3.1%	12.3%	6.0%	2.1%
<i>rrs</i>	13.2%	5.7%	12.5%	5.7%	3.5%	0%
<i>gidB</i>	13.2%	25.0%	21.8%	12.7%	2.7%	2.1%
Reference	This study	[27]	[28]	[29]	[30]	[31]

Table 5

Distribution of mutations in *gidB* among different genotypes of 197 streptomycin-resistant *Mycobacterium tuberculosis* isolates.

<i>gidB</i> nucleotide (amino acid substitution)	Lineage 1	Lineage 2	Lineage 3	Lineage 4	No. of isolates
G28C (A10P)			2		2
G28C (A10P), A119G (E40G)	1				1
C56G (A19G)*, γ, θ				1	1
G59C (A20P)*, β	1				1
G79C (A27P)*			1		1
C80T (A27V)*, α		1			1
G106C (V36L)*			1		1
C142A (H48 N), γ				2	2
T236G (L79Y)			1		1
G244C (A82P)*			1		1
T302G (L101E)*			1		1
T335G (V112G)*			1		1
T371G (V124G)*			1		1
C398T (A133V)*, α		1			1
C413T (A138V), γ				1	1
G471A (G157G)*, α		2			2
A487G (K163E)			1		1
C492T (G164G)*			1		1
T583C (Y195H), α		7			7
G630A (G210G)*, α		1			1
A652G (M218V)*, γ				2	2
243insGC (Fs)*, γ				1	1
98delG (R33Fs)			6		6
98delG (R33Fs), β	3				3
98delG (R33Fs), γ				3	3
112delC (P38Fs)*, β	1				1
112delC (P38Fs)*, γ				1	1
254delA (D85Fs)*, γ				1	1
347delG (R116Fs)			1		1
372delG (V124Fs)*			1		1
384delG (L128Fs)			2		2
A	1	104			105
B	4				4
Γ				8	8
γ, θ				3	3
–			26		26
Total	11	116	47	23	197

A276C (E92D) = α, G330T (V110V) = β, G615A (A205A) = γ and T47G (L16R) = θ

Fs denotes frameshift mutation.

No. represents number.

* denotes mutation previously not reported.

- denotes no mutation.

Del denote deletion mutation.

C789T [37]; and *rpsL* K88Q/*rrs* A905G and *rpsL* K43R/*rrs* A513C [38] have been reported. In contrast, Jagielski et al., had reported that mutations in *rrs* and *rpsL* were mutually exclusive, that is isolates with *rrs* mutations had no *rpsL* mutations and vice versa [39]. The discrepancy in our results might be due to the presence of mixed population of *rpsL* K43R mutants and *rrs* A907C mutants. This was confirmed to be 1:1 mixture of A and G at nucleotide position 128 in *rpsL* by the electropherogram obtained. One strain harbored mutations in *rpsL* causing K88R and *rrs* A644G. One studies had reported an isolate with mutation in *rpsL* K43R and a second mutation in *rrs* nucleotide 645 A deletion [26]. A mutation in *rrs* A644G is a novel mutation and this mutation is located far away from hot spot mutation regions of *rrs*, the 530 loop and 912 region, which is associated with the STR-resistance in MTB. As mutation in *rpsL* causing K88R is enough to cause STR-resistance in MTB, a mutation in *rrs* A644G might not be associated with the STR-resistance in MTB but be a natural polymorphism. Other strain harbored mutations in *rpsL* causing G118D and *rrs* A514C. As MTB has a single rRNA operon [40], a single mutation in *rrs* A514C is enough to cause STR-resistance in MTB. Mutation G118D have not been previously reported and this mutation is located far away from frequent mutations

in *rpsL*, K43R and K88R, which cause STR-resistance in MTB. It is unclear that the mutation in *rrs* A644G and mutation in *rpsL* causing G118D may affect the mechanism of STR-resistance in MTB. So, further studies should be carried out to find out the exact role of these mutations in causing STR-resistance in MTB.

We observed that 13.7% (27/197) of STR-resistant isolates showed many mis-sense mutations in *gidB* that were close to active or part of the interacting amino acid residues between the *GidB* protein and S-adenosylmethionine (SAM) which are critical for methylation at the position G517 of 16S rRNA [27]. These alterations may have an impact on the function of *GidB* which leads to low affinity of STR to 16S rRNA and hence contribute to STR-resistance in MTB. The missense mutations in *gidB* codons causing amino acid substitutions A19G, R20P, A27P, A27V, V36L, A82P, L101E, V112G, V124G, A133V, M218V.

The *gidB* mutants harboring co-mutations in *rpsL*/*rrs* genes were found in 12.7% (25/197) of isolates. These co-mutations of in *rpsL*/*rrs* of *gidB* mutants might suggest the evolution of *rpsL*/*rrs* mutations for its existence as a compensation for losing methylation function of *GidB* caused by distortion of its structure and consequently results in high level resistance to STR [10,36].

It was found that 13.2% (26/197) of isolates showed mutations in *gidB* without those in *rpsL* and *rrs* genes. All 13 mis-sense mutations had an exclusive *gidB* mutation without mutations in *rpsL* and *rrs* conferring these *gidB* mutations contribute to STR-resistance in MTB. Deletion mutations in *gidB* nucleotide without those in *rpsL* and *rrs* (112delC, 254delA, 347delG and 372delG) and insertion mutations in *gidB* nucleotide (243insGC) causing frameshift in *gidB* in STR-resistant isolates found in this study which alter the *GidB* protein and consequently leading to low-level STR-resistance in MTB. Deletions with variable sizes at different positions in *gidB* were also reported [8,10,30]. Frameshift mutations (112delC, 254delA, 347delG, 372delG, 384delG and 243insGC) were previously reported. The deletion mutation in *gidB* delG at nucleotide 98 have also been reported in both STR-susceptible and STR-resistant isolates [39,41]. Mutations in *gidB* found exclusively in STR-resistant strains play a role in STR-resistance in MTB. These 13 mis-sense mutations and 5 frameshift mutations in *gidB* were highly associated with STR-resistance in MTB.

There were four *gidB* polymorphisms phylogenetically specific to MTB lineages which were found in 75.6% (149/197) of isolates. The missense mutation E92D conferred by 100% (116/116) of lineage 2 isolates which was significantly associated with the Beijing family of lineage 2 ($P < 0.001$). Similar indication of high specificity to Beijing strains have been reported in previous studies [34–36,42]. The silent mutation in *gidB* codon at V110 was significantly associated with EAI strains ($P < 0.001$), which is also supported by previous findings [34, 42]. Another silent mutation in *gidB* at codon A205 was present in 100% (23/23) of the Euro-American strains of lineage 4 which was significant ($P < 0.001$). In contrast, polymorphisms in *gidB* codon at A205 was absent from almost all of the Euro-American strains in which this allele may be used as a specific marker for differentiating Euro-American strains from non-Euro-American strains [34,42]. Finally, all LAM strains conferred the allele mutation in *gidB* codon at L16 was significant ($P < 0.01$) which is also confirmed by the previous findings [36,43,44].

In total, 94.4% (186/197) of STR-resistant isolates harbored mutations in either *rpsL* or *rrs* or *gidB* alone or in combinations. This suggests that mutations in *rpsL*, *rrs* and *gidB* play a significant role in the mechanism of STR-resistance in MTB. In this study, we found 11 isolates (one lineage 1 strains, two lineage 2 strains, seven lineage 3 strains and one lineage 4 strains) do not harbor mutation in any of the genes studied, accounting for 5.6% of the STR-resistant strains. This suggests that there might be an involvement of other genes in the mechanism of STR-resistance such as the efflux system, cell membrane permeability barrier or aminoglycosides converting enzymes.

In conclusion, the mutations in *rpsL* and *rrs*; and some exclusive mutations in *gidB* identified in this study are responsible for STR-resistance in MTB. The sequence analysis of mutations in *rpsL*, *rrs* and

gidB could be used for the establishment of rapid and effective molecular diagnostic tools to detect STR-resistant MTB strains in Nepal. This study can be further implicated for designing rapid diagnostic tools considering geographic specificity in which MDR-TB transmission can be interrupt worldwide.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.tube.2020.101985>.

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